SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 9

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Task Description -1 (Documentation – Function Summary Generation)
Task:
Use AI to generate concise functional summaries for each Python function in a given script.
Instructions:
- **Provide a Python script to the AI.**
- **Ask the AI to write a short summary describing the purpose of each function.**
- **Ensure summaries are brief and technically accurate.**
- **Do not include code implementation details.**

Expected Output -1:
A Python script where each function contains a clear and concise summary explaining its purpose.

Prompt:-
Write a Python program that calculates the factorial of a number using an iterative method (without creating extra helper functions). Also, include a short and clear docstring that explains what the factorial logic does.
Code:-

```python
#Write a Python program that calculates the factorial of a number using an iterat
def factorial(n):

    if n < 0:
        raise ValueError("Factorial is not defined for negative numbers.")
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result
# Test cases
print(factorial(0))  # Output: 1
print(factorial(1))  # Output: 1
print(factorial(5))  # Output: 120

try:
    print(factorial(-1)) # This should raise an error
except ValueError as e:
    print(e)  # Output: Factorial is not defined for negative numbers.
```

Output:-

```
PS C:\Users\Dell\Desktop\PYTHON> & C:/Users/Del
120
Factorial is not defined for negative numbers.
```

Justification:-

The docstring clearly explains that the program calculates the factorial of a given number and returns the result. It also mentions how special cases like 0 and 1 are handled. This makes it easier for anyone reading the code to quickly understand its purpose without going through every line. Good documentation helps improve clarity and makes the program easier to maintain in the future.

Task Description -2 (Documentation – Logical Explanation for Conditions and Loops)
Task:
Use AI to document the logic behind conditional statements and loops in a Python program.
Instructions:
- **Provide a Python program without comments.**
- **Instruct AI to explain only decision-making logic and loop behavior.**
- **Skip basic syntax explanations.**

Expected Output -2:
Python code with clear explanations describing the logic of conditions and loops.

Prompt:-
write a code for a logic of conditional statements and loops to calculate the prime factors of a number and write a function docstring concise summary and only decision making logic and loops behaviour, skip basic syntax explanation.

Code:-

```python
#Create a Python program to find the prime factors o
def prime_factors(n):
    """
    factors = []
    factor = 2
    while n > 1:
        if n % factor == 0:
            factors.append(factor)
            n //= factor
        else:
            factor += 1
    return factors
# Test cases
print(prime_factors(28))
print(prime_factors(18))
print(prime_factors(29))
```

Output:-

```
PS C:\Users\Dell\Desktop\PYTHON>
[2, 3, 3]
[29]
```

Justification:-
The prime_factors function uses logical conditions and loops to identify all the prime factors of a number. First, it checks how many times the number is divisible by 2. After that, it checks other possible factors starting from 3 up to the square root of the number. The loop keeps running until all factors are found. This approach avoids unnecessary checks and makes the program efficient.

Task Description -3 (Documentation – File-Level Overview)
Task:
Use AI to generate a high-level overview describing the functionality of an entire Python file.
Instructions:
• Provide the complete Python file to AI.
• Ask AI to write a brief overview summarizing the file's purpose and functionality.
• Place the overview at the top of the file.
Expected Output -3:
A Python file with a clear and concise file-level overview at the beginning.

Prompt:-
generate a complete python file below clear, concise file-level overview and insert it at the very top as a module docstring.

Code:-

```python
def factorial(n):
    """Calculate the factorial of a number n and return the result."""
    if n < 0:
        return "Factorial is not defined for negative numbers."
    elif n == 0 or n == 1:
        return 1
    else:
        result = 1
        for i in range(2, n + 1):
            result *= i
        return result
def prime_factors(n):
    """Calculate the prime factors of a number n and return them as a list."""
    factors = []
    # Check for number of 2s that divide n
    while n % 2 == 0:
        factors.append(2)
        n //= 2
    # n must be odd at this point, so we can skip even numbers
    for i in range(3, int(n**0.5) + 1, 2):
        while n % i == 0:
            factors.append(i)
            n //= i
    # This condition is to check if n is a prime number greater than 2
    if n > 2:
        factors.append(n)
    return factors
# Example usage
number = 6
print(f"The factorial of {number} is {factorial(number)}")
number = 28
print(f"The prime factors of {number} are {prime_factors(number)}")
```

Output:-

```
The factorial of 6 is 720
The prime factors of 28 are [2, 2, 7]
```

Justification:-
The module docstring provides a clear and concise overview of the program's functionality, describing the purpose of each function and how they operate. It also mentions the inclusion of example usage, which helps users understand how to utilize the functions effectively. This level of documentation enhances readability and makes it easier for other developers to understand and reuse the code in their projects.

Task Description -4 (Documentation – Refine Existing Documentation)
Task:
Use AI to improve clarity and consistency of existing documentation in Python code.
Instructions:

• Ask AI to rewrite the documentation to improve clarity and consistency.
• Ensure technical meaning remains unchanged.

Expected Output -4:
Python code with refined and improved documentation that is clear and consistent.

Prompt:-
write a program to improve clarity and consistency of documentation in python code, containing basic or unclear comments.

Code:-

```python
# task-4
# write a program to improve clarity and consistency of documentation in python
"""This module provides functions to calculate the factorial of a number and to
"""
def factorial(n):
    """Calculate the factorial of a number n and return the result."""
    if n < 0:
        return "Factorial is not defined for negative numbers."
    elif n == 0 or n == 1:
        return 1
    else:
        result = 1
        for i in range(2, n + 1):
            result *= i
        return result
def prime_factors(n):
    """Calculate the prime factors of a number n and return them as a list."""
    factors = []
    # Check for number of 2s that divide n
    while n % 2 == 0:
        factors.append(2)
        n //= 2
    # n must be odd at this point, so we can skip even numbers
    for i in range(3, int(n**0.5) + 1, 2):
        while n % i == 0:
            factors.append(i)
            n //= i
    # This condition is to check if n is a prime number greater than 2
    if n > 2:
        factors.append(n)
    return factors
# Example usage
number = 9
print(f"The factorial of {number} is {factorial(number)}")
number = 56
print(f"The prime factors of {number} are {prime_factors(number)}")
```

Output:-

```
The factorial of 9 is 362880
The prime factors of 56 are [2, 2, 2, 7]
```

Justification:-
The module docstring provides a clear and concise overview of the functionality offered by the code, explaining what each function does and how they can be used. The function docstrings are also clear and informative, detailing the purpose of each function, the expected input, and the output. This level of documentation enhances readability and makes it easier for other developers to understand and utilize the code effectively.

Task Description -5 (Documentation – Prompt Detail Impact Study)
Task:
Study the impact of prompt detail on AI-generated documentation quality.

Instructions:
Create two prompts: one brief and one detailed.

- **Use both prompts to document the same Python function.**
- **Compare the generated outputs.**

Expected Output -5:
A comparison table highlighting differences in completeness, clarity, and accuracy of documentation.

Prompt:-
Generate two docstrings for the same Python function—one from a very short prompt and one from a detailed prompt—then compare them in a table for completeness, clarity, and accuracy, using natural, human-like language throughout.

Code:-

```
# task-5
# Generate two docstrings for the same Python function-one from a very short prompt and one from a detailed prompt-then compare them in a table for completeness, clarity, and accuracy, using natural, human-like l
def calculate_area(radius):
    """Calculate the area of a circle given its radius."""
    import math
    if radius < 0:
        return "Radius cannot be negative."
    return math.pi * radius ** 2
# Short prompt docstring
"""Calculate the area of a circle given its radius.
"""
# Detailed prompt docstring
"""Calculate the area of a circle given its radius.
This function takes a single argument, radius, which represents the distance from the center of the circle to its edge.
The function checks if the radius is negative and returns an error message if it is.
If the radius is valid, it uses the formula for the area of a circle (A = πr²) to compute and return the area.
"""
# Comparison table
comparison_table = """
| Aspect       | Short Prompt Docstring                                    | Detailed Prompt Docstring                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---|---|---|
| Completeness | Provides a basic description of the function's purpose. | Offers a comprehensive explanation of the function's behavior, including input validation and the formula used. |
| Clarity | Clear but lacks detail on input validation and the formula. | Very clear, with detailed information on what the function does, how it handles invalid input, and the mathematical formula used. |
| Accuracy | Accurate in describing the function's purpose. | Accurate and provides a complete understanding of the function's behavior and expected input. |
"""
print(comparison_table)
```

Output:-

```
| Aspect       | Short Prompt Docstring                                    | Detailed Prompt Docstring                                                                                                                                                                                                                                                                                                                                
|---|---|---|
| Completeness | Provides a basic description of the function's purpose. | Offers a comprehensive explanation of the function's behavior, including input validation and the formula used. |
| Clarity | Clear but lacks detail on input validation and the formula. | Very clear, with detailed information on what the function does, how it handles invalid input, and the mathematical formula used. |
| Accuracy | Accurate in describing the function's purpose. | Accurate and provides a complete understanding of the function's behavior and expected input. |
```

Justification:-
The detailed prompt docstring is more complete, providing not only the purpose of the function but also details about input validation and the mathematical formula used. This makes it clearer and more informative for users who may want to understand how the function works or modify it in the future. The short prompt docstring, while accurate, lacks this depth of information, which can lead to confusion or misuse of the function.